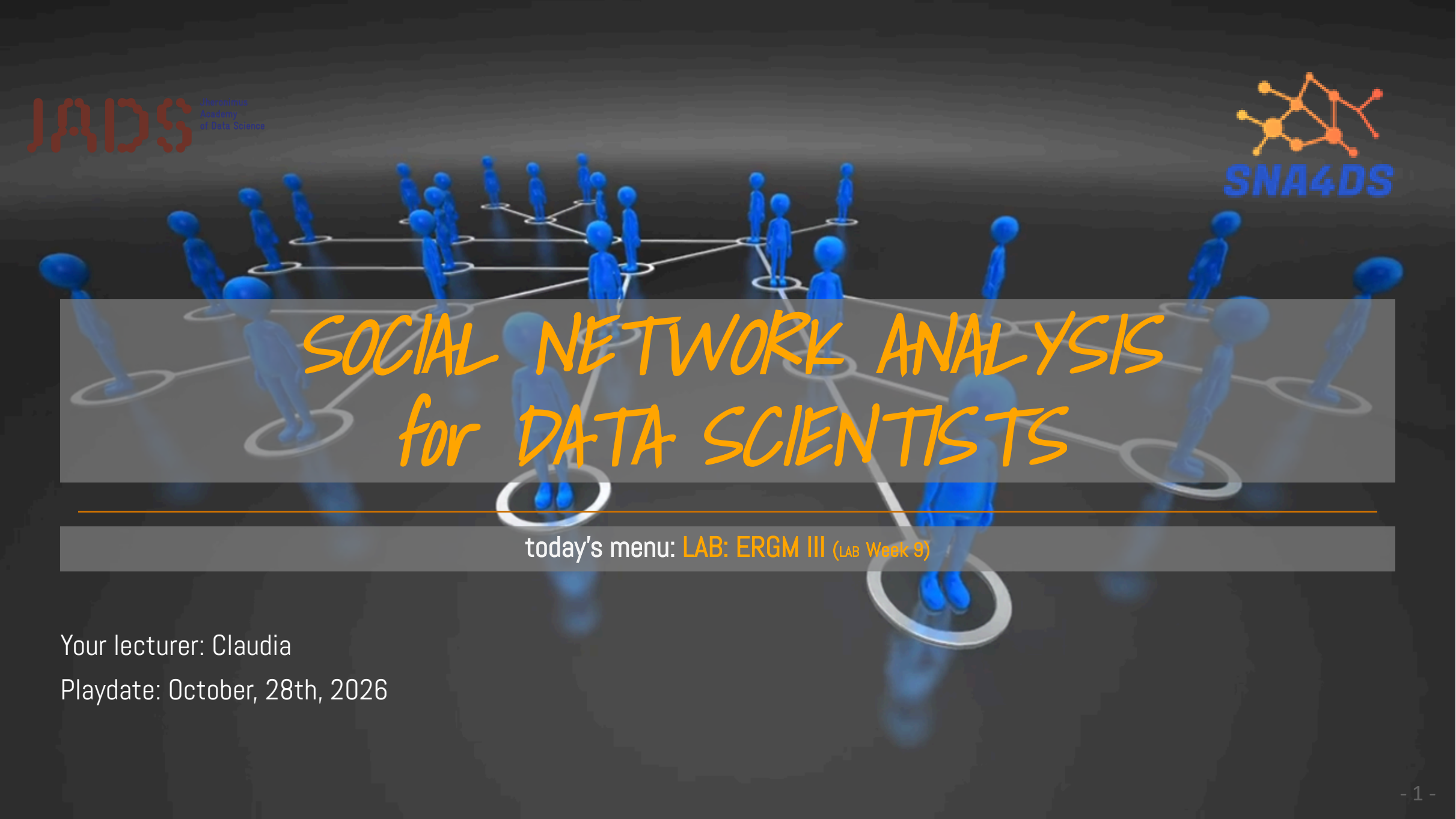


SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: LAB: ERGM III (LAB Week 9)

Your lecturer: Claudia

Playdate: October, 28th, 2026



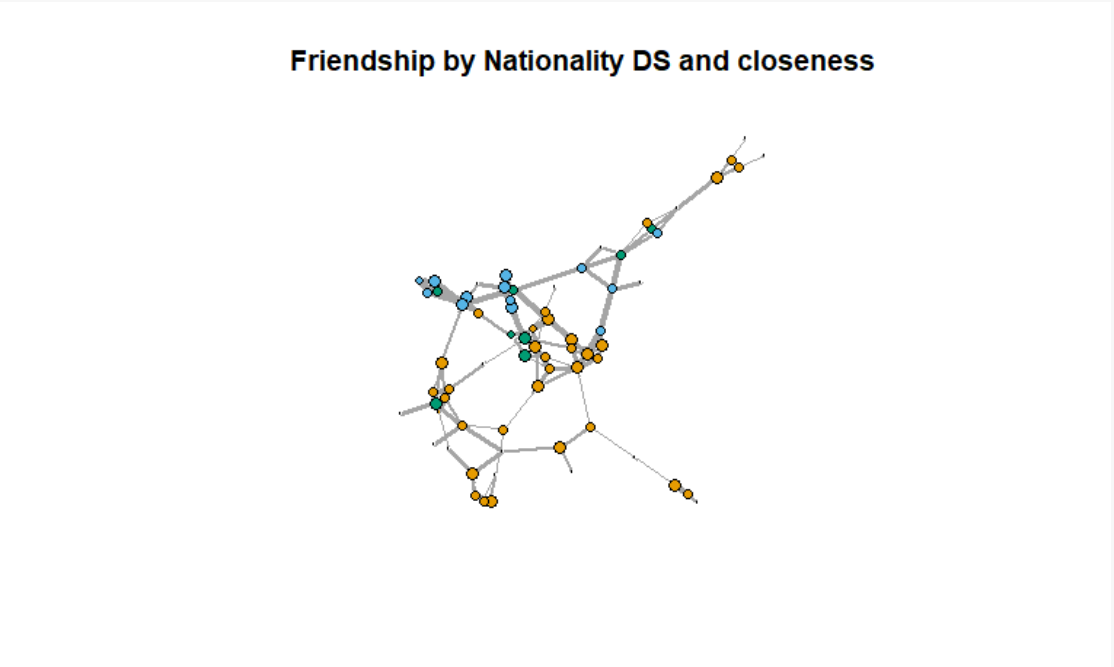
Credit

Menu

- Super Lab - Real Time ERGM!
- Final Recommendations
- ERGMs Qs & As

Real Time ERGM

Let's Research JADS Data! :)




```
setwd("C:/Users/...")
```

```
load("JADSmaster2022.Rdata")
```

80:00

Final Recomendations

Remember

- Network outcome variable
- Always use endogenous terms
- Handling large networks
- Handling complex cases
- Seven step flow approach

Network outcome variable

You already know that ERGMs work with a network outcome variable.

Make sure you are developing your theory accordingly!

Some examples of Hypotheses

- It is more likely that people of the same age are friends
- More popular scholars prefer to collaborate with other popular scholars
- Companies want to trade with other companies in the same sector

Thinking in Varibles:

It is more likely that people of the same age are friends

- outcome: friendship
- predictor: age (numeric)

More popular scholars prefer to collaborate with other popular scholars

- outcome: collaboration
- predictor: popularity (numeric/ordinal)

Companies want to trade with other companies in the same sector

- outcome: trade
- predictor: sector (categorical)

Always use endogenous terms -

Even if your hypotheses do not point you to endogenous terms, they still must be in the model (2 H at least).

Two reasons

- Otherwise it is just like a logistic regression
- The model won't work well
- It's compulsory

ERGMs explain relationships.

Allow them to do their job

Handling large networks

Ideally, for this assignment, you should stay below 500 nodes

Handling large networks

Ideally, for this assignment, you should stay below 500 nodes

Why?

Handling large networks

SNA4DS

Ideally, for this assignment, you should stay below 500 nodes

Why?

Just because it takes too much time to run these models and you have a tight deadline.

Handling large networks

If your project requires large networks (even around 500)

1. use super efficiently control ergm (always use it) to optimize the simulation parameters
2. contraint the model so that the simulation gets a little faster
3. run it on a server
4. parallelize

Handling complex cases

Some of you are re-constructing fairly complex networks

Make sure that you know what the terms you insert in the model mean

It is better to have a simple project than super fancy attempts that nobody can understand.

Also, make sure you don't do too many transformations with your data before putting them into the model.

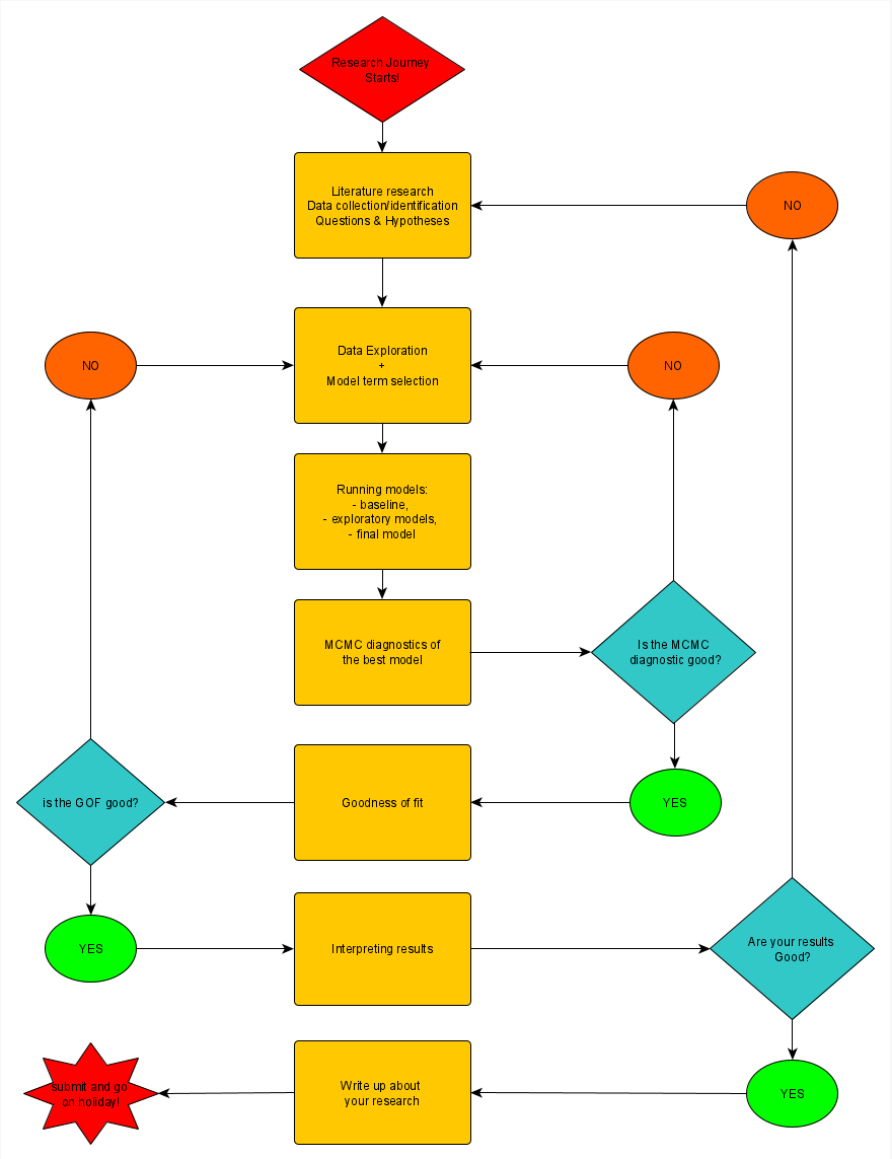
SUPER DISCLAIMER

WE DON'T CARE ABOUT STATISTICAL SIGNIFICANCE

- Significant effects + good MCMC diagnostics + good GOF = AWESOME

BUT

- Significant estimates + bad MCMC diagnostics + bad GOF = bad mark
- NON-Significant estimates + good MCMC diagnostics + good GOF = good mark (IF YOU CAN EXPLAIN WHY)


Research Journey Starts!
Literature research
Data collection/identification
Questions & Hypotheses
NO
Data Exploration
+
Model term selection
NO
NO
Running models:
- baseline,
- exploratory models,
- final model
MCMC diagnostics of the best model
Is the MCMC diagnostic good?
is the GOF good?
Goodness of fit
YES
YES
Interpreting results
Are your results Good?
submit and go on holiday!
Write up about your research
YES

ERGMs Qs & As

Final point: Think carefully about your research questions!



Good Luck backing your Projects!

